

Theme #1 - Multi-Agent Interactions

Environments for this theme involve cooperation, competition, negotiation, and coalition formation. Learning from these environments will enable agents to model the beliefs and incentives of others in partially observable settings. This drives theory-of-mind reasoning and emergent strategic behavior.

Expected Outcome: an environment that can be used to train multi-agent task handling in a LLM

Example environments: Market simulations, compute-allocation negotiations, collaborative puzzle worlds, mixed cooperative/competitive strategy games.

Sub-themes with bonus prizes.

- **Fleet AI.** Scalable Oversight: Environments that train oversight agents to monitor, analyze, and explain the behavior of other AI agents operating in complex, multi-agent settings.
- **Halluminate.** Multi-Actor Environments: Build a realistic environment where an agent interacts with and manages multiple actors (agents) to discover and achieve the task

Theme #2 - (Super) Long-Horizon Planning & Instruction Following

You will build environments that require deep, multi-step reasoning with sparse or delayed rewards. After using these environments, the goal is to enable agents to decompose goals, track state over extended trajectories, and recover from early mistakes. The aim is to push beyond shallow next-token reasoning toward structured planning and durable internal representations.

Expected Outcome: an environment that can capture and improve LLM behaviour on challenging long horizon tasks that need long running sessions beyond context memory limits.

Example environments: Research-planning simulators, large-scale codebase refactoring tasks, strategic resource management worlds, long-horizon logistics optimization, extremely complicated long-horizon instruction following (e.g., 300 instructions scattered around).

Sub-themes with bonus prizes.

- **Scale AI.** Environments for long horizon workflows for non-code use cases within a business setting: focusing on either Sales, Project management, or HR & IT.
- **Mercor.** Make an environment with capped/uncapped rewards where frontier model rewards scale with token output.

Theme #3 - World Modeling

#3.1 Professional Tasks

Here you will develop environments that require real interaction with tools, APIs, or dynamic systems where the model is expected to do real hard work instead of exploiting short-cuts to arrive at the desired outcome. Learning from these environments will enable agents to maintain consistent internal state, update beliefs based on outcomes, and orchestrate multi-step workflows. The goal is to strengthen causal reasoning and persistent world models.

Expected Outcome: an environment capturing nuances of a defined partially observable world and improve LLM interaction with it

Example environments: Dynamic browser/API ecosystems, enterprise applications, scientific workflow loops (papers → code → experiments), economic simulations with feedback, tool-discovery benchmarks.

Sub-themes with bonus prizes.

- **Scaler AI Labs.** Multi-App RL Environment for Enterprise Workflows: Create RL environments to demonstrate complex workflows, business rule nuances etc in a large enterprise

#3.2 Personalized Tasks

Here we will develop an environment that offers real personalized task handling, imagine replying to personal messages or handling dinner conflicts due to work conflicts, replying to tough emails. Think any personal assistant tasks

Expected Outcome: An environment that gives the model a realistic simulation of handling personal tasks, conflicts and managing them as delegations

Example environments: Executive Assistant Meeting Planner, Dinner and drive planning, email and message replying, shopping, etc

Sub-themes with bonus prizes.

- **Patronus AI.** Consumer Workflows with Schema Drift: Multi-step consumer workflow environments where the underlying data schemas, API contracts, and t&cs/policies/rules change.

Theme #4 - Self-Improvement

The focus here is to create environments where agents can learn to generate new challenges, escalate difficulty, and improve through self-play or adaptive curricula. Rather than optimizing fixed tasks, the goal is for agents to learn to drive their own capability growth. The objective is recursive skill amplification.

Expected Outcome: an environment for improving self-play of a LLM over a defined set of tasks

Example environments: Self-play negotiation arenas, auto-generated math/proof tasks, evolving coding competitions, adaptive RL curricula.

Sub-themes with bonus prizes.

- **Snorkel AI.** Simulated Experts-in-the-Loop: Environment that simulates interactions with real subject-matter experts, with changing requirements / preferences.

Theme #5: Wild Card - Impress Us!

We do not want to limit your focus if your idea doesn't fit the boxes above, we want and WILL reward out of box tasks, please be creative but remember to add submissions that meaningfully add value to LLM training on a certain task.

Guidelines for Problem Statement

- It is **NOT** mandatory to choose the same problem statement as Round 1. Only choose the same problem statement if it aligns with the above provided Hackathon themes.
- You can start working on your problem statement once you have finalized it. Post-training can be done onsite on 25th & 26th when you receive compute credits for HuggingFace.
- Before the onsite, we suggest you work on building the environment, agent behaviours, reward model and evaluate if your work aligns with the [judging criteria](#) given below.

Judging Criteria

Minimum requirements:

- Usage of OpenEnv (latest release)
- Show a minimal training script for your environment using Unsloth or HF TRL in Colab
- Write a mini-blog on HuggingFace or mini-video on YouTube talking about your submission, <2 minutes
- Your OpenEnv compliant environment should be hosted on Hugging Face Spaces.

First Round Judging Overview

- **Pitch Format:** Each team has 3 minutes to pitch, followed by 2 minutes for Q&A (5 minutes total).
- **Evaluation:** Teams will be scored based on the following criteria:
 1. **Environment Innovation (40%):** Is the environment novel, creative, or challenging? Does it meaningfully test the agent's behavior?
 2. **Storytelling (30%):** Does the team clearly explain the problem, environment, and agent behavior? Is the demo engaging and easy to follow?
 3. **Showing Improvement in Rewards (20%):** Does the demo provide observable evidence of training progress (reward curves, metrics, or before/after behavior)?
 4. **Reward and Training Script/Pipeline Setup (10%):** Is the reward logic coherent, and does the pipeline produce meaningful improvement in the agent's inference (how it acts in the environment)?

Final Note

Judges are looking for environments that push the frontier of what we can train LLMs to do. Be

ambitious. Pick a problem you find genuinely interesting; that almost always produces better

work than chasing what you think judges want. Good luck.